
MORTGAGE DELINQUENCY RISK ANALYSIS

Technical Documentation

Dashboard Design | Data Architecture | Modelling Decisions | Interview Reference

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This document explains the technical decisions, design reasoning, and analytical choices behind the Mortgage Delinquency Risk Analysis project. It is intended as the definitive technical reference for the project and as a structured interview preparation resource.

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1 INTRODUCTION

Purpose of This Document

This Technical Documentation is the third and final major document in the Mortgage Delinquency Risk Analysis project documentation suite. Where the Executive Credit Risk Report communicates findings to senior management, and the Business Case Study explains the complete analytical journey for portfolio and career audiences, this document explains the technical reasoning behind every significant decision made during the project's construction.

The primary purpose of this document is twofold. First, it serves as a technical reference manual — the source of record for every architectural, modelling, and design decision that shaped the project. Second, it serves as a structured interview preparation resource — enabling confident, precise defence of technical choices during credit risk analytics interviews.

Document Relationships

Document	Primary Audience	Core Question Answered
Executive Credit Risk Report	CRO, Credit Risk Committee, Senior Management	What did we find, and what should management do?
Business Case Study	Recruiters, Hiring Managers, Portfolio Reviewers, Analysts	How was the project built, and what is its business value?
Technical Documentation (this document)	Technical Interviewers, Senior Analysts, Future Maintainers	Why were specific technical decisions made, and how do they hold up under scrutiny?
Power BI Dashboard	Executive Users, Risk Managers, Portfolio Monitors	What is the current portfolio risk status across six analytical dimensions?

What This Document Does Not Repeat

This document does not reproduce the findings, business narratives, or recommendations already documented in the Executive Report and Business Case Study. Readers seeking portfolio health metrics, borrower risk findings, geographic analysis conclusions, or strategic recommendations should refer to those documents. This document addresses the technical layer beneath those outputs: the choices that produced them.

Navigation note: Sections 2 and 3 cover the project architecture and data modelling philosophy. Section 4 explains dashboard design principles. Section 5 is the core technical walkthrough — the most interview-relevant section. Sections 6–9 address specific technical decisions, limitations, future enhancements, and lessons learned.

2 SOLUTION ARCHITECTURE

The project was built across five distinct architectural layers, each serving a specific analytical purpose. Understanding the complete pipeline — from raw file ingestion through to executive dashboard output — is essential for understanding why individual technical decisions were made.

Figure 1. Solution Architecture

End-to-end analytical workflow from raw mortgage servicing data to executive decision support.

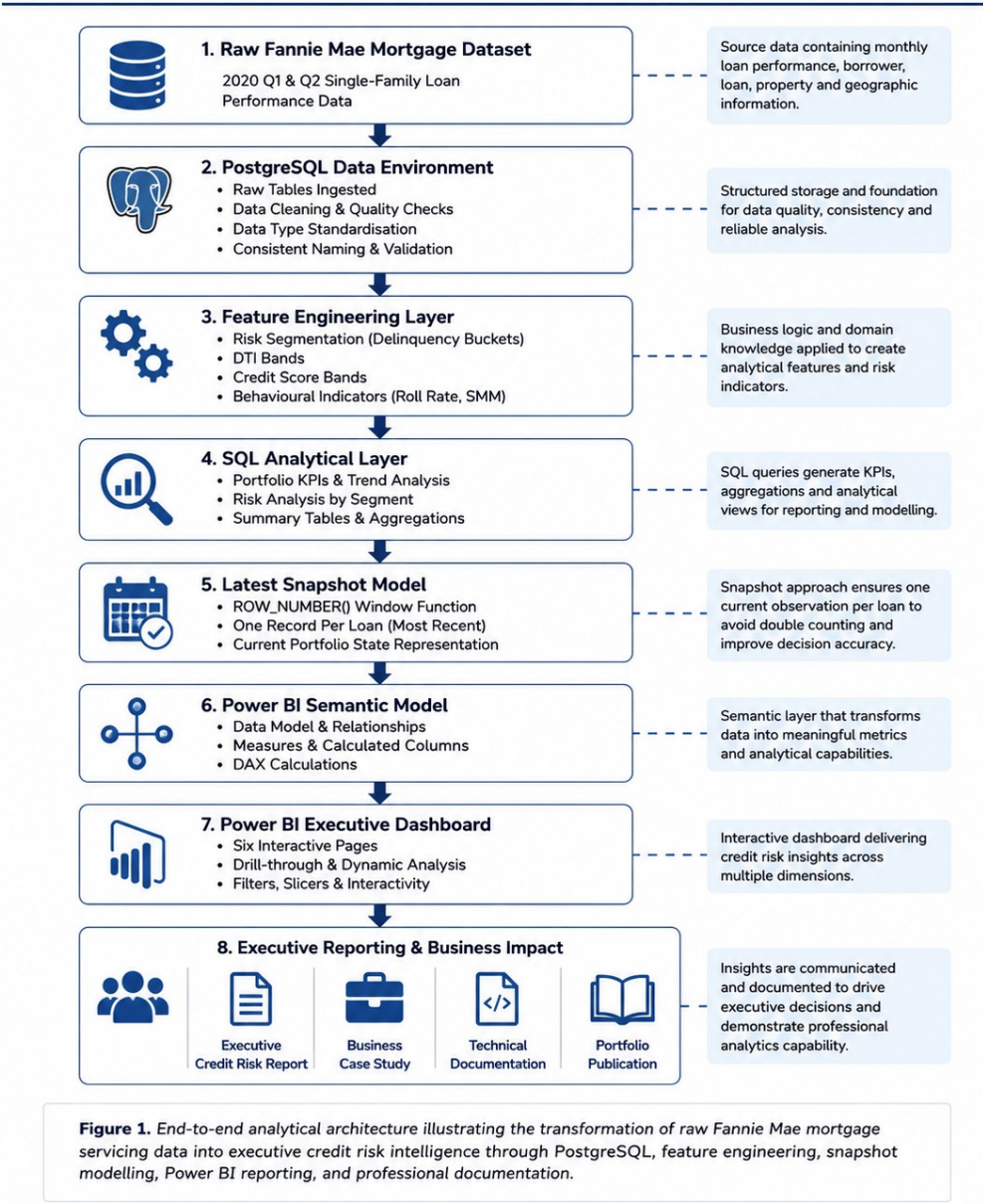


Figure 1. Solution Architecture

Eight-layer end-to-end analytical architecture: from raw Fannie Mae source data through PostgreSQL ingestion, data cleaning, feature engineering, SQL analytical layer, latest-snapshot model, Power BI semantic model, six-page executive dashboard, and professional documentation outputs.

End-to-end pipeline diagram: Raw CSV files → PostgreSQL ingestion → Cleaning view → Typed view → Feature-engineered view → Summary tables → Latest-snapshot fact table → Power BI model → Six-page dashboard → Documentation ecosystem.

Layer	Technology	Object(s)	Purpose
1 — Raw Ingestion	PostgreSQL	mortgage_raw_q1, mortgage_raw_q2	Load pipe-delimited source files. All 108 columns imported as TEXT to guarantee successful ingestion without type errors.
2 — Data Cleaning	PostgreSQL View	mortgage_clean_v	Remove 40,190 records with null origination credit score. Filter applied once at this layer; downstream objects inherit the clean dataset.
3 — Type Standardisation	PostgreSQL View	mortgage_typed_v	Cast analytical fields to correct data types (INTEGER, NUMERIC, TEXT). Categorical codes and date strings retained as TEXT.
4 — Feature Engineering	PostgreSQL View	mortgage_fe_v	Derive Credit Score Band, LTV Band, DTI Band, Risk Factor Count, Risk Segment, Delinquency Flag, Serious Delinquency Flag. Primary analytical source.
5 — Summary Tables	PostgreSQL Tables	Multiple pre-aggregated tables	Pre-calculate segment-level KPIs to avoid repeated full-scan aggregation against 84M rows during EDA.
6 — Snapshot Table	PostgreSQL Table	Latest-snapshot fact table	One row per unique loan (most recent observation). Approximately 1.92M records. Primary Power BI source.
7 — Power BI Model	Power BI Desktop	Data model + DAX measures	Import snapshot table. Build DAX measures for all KPIs. Apply REMOVEFILTERS where required for consistent KPI strip.
8 — Dashboard	Power BI	Six-page report	Six executive pages covering portfolio health, borrower risk, structural risk, geographic risk, behavioural risk, and surveillance.
9 — Documentation	Word + PDF	Three documents	Executive Report, Business Case Study, Technical Documentation (this document).

Source Dataset Characteristics

The Fannie Mae Single-Family Loan Performance dataset arrives as pipe-delimited text files with no column header row. Fields are identified by their positional index (Field Position 1 through 113 in the Fannie Mae glossary). The CSV structure introduces a leading pipe character, meaning CSV Column 1 is always blank, and glossary Field Position N maps to CSV Column N+1. This offset was identified during the field mapping exercise and documented for all downstream query work.

TECHNICAL NOTE 108 columns were confirmed in the CSV structure (Column1 through Column108), compared to 113 positions in the Fannie Mae glossary. The difference reflects fields introduced in later dataset versions not yet present in the 2020 Q1–Q2 extract. All 108 present columns were mapped and classified before any analytical work began.

PostgreSQL as the Analytical Engine

PostgreSQL was selected as the analytical engine for three reasons. First, the dataset scale — 84 million rows — exceeds the practical limits of Excel-based analysis and is well within PostgreSQL's optimised set-based processing capabilities. Second, the view-based architecture allows changes to be propagated

downstream automatically: modifying the cleaning logic in `mortgage_clean_v` immediately affects all objects built on top of it without requiring manual refresh. Third, SQL is the lingua franca of credit risk analytics in financial institutions — documenting the analysis in SQL produces directly transferable portfolio evidence.

3 DATA MODELLING PHILOSOPHY

The Original Dataset — Monthly Servicing Panel

The source dataset is a loan-month panel: each unique loan generates one row per calendar month it exists in the portfolio. A loan originated in January 2020 and still active in June 2020 contributes six rows to the dataset — one for each monthly reporting period. This grain is explicitly designed for longitudinal performance analysis: it enables tracking of how individual loans transition between delinquency stages over time, which is the analytical foundation for roll rate and cure rate calculation.

The origination attributes for each loan — credit score, LTV, DTI, loan purpose, property type — are static. They are identical across every monthly row for the same loan. The performance attributes — delinquency status, current UPB, modification flag — are dynamic. They update each month to reflect the loan's evolving status.

KEY INSIGHT *The panel structure is correct for time-series analysis and incorrect for current-state portfolio reporting. Using panel grain for executive KPIs without transformation produces metrics that count each loan multiple times — once per month it has existed in the portfolio. This is the problem the snapshot model solves.*

The Grain Problem — Why Panel Data Misleads Executive Reporting

Consider the total exposure calculation. Summing Current UPB across all 84.18 million rows produces approximately \$19.97 trillion. This figure is not wrong — it is the cumulative sum of all monthly outstanding balances across the portfolio's full observation history. But it is not what an executive means when they ask 'how much exposure do we have?' They are asking about the current outstanding balance of the portfolio today — not the sum of all historical monthly balances.

The same distortion affects every current-state metric. A delinquency rate calculated on the full panel includes loans that were delinquent at some point in their history but have since cured — overstating current credit stress. A loan count from the panel counts each loan once per month of observation — 12 months of history for a loan means it is counted 12 times. None of these are errors in the data; they are consequences of using the wrong grain for the wrong question.

Figure 2. Snapshot Model Architecture (Latest Snapshot Approach)

One record per loan representing the most recent state to enable accurate portfolio-level risk analysis.

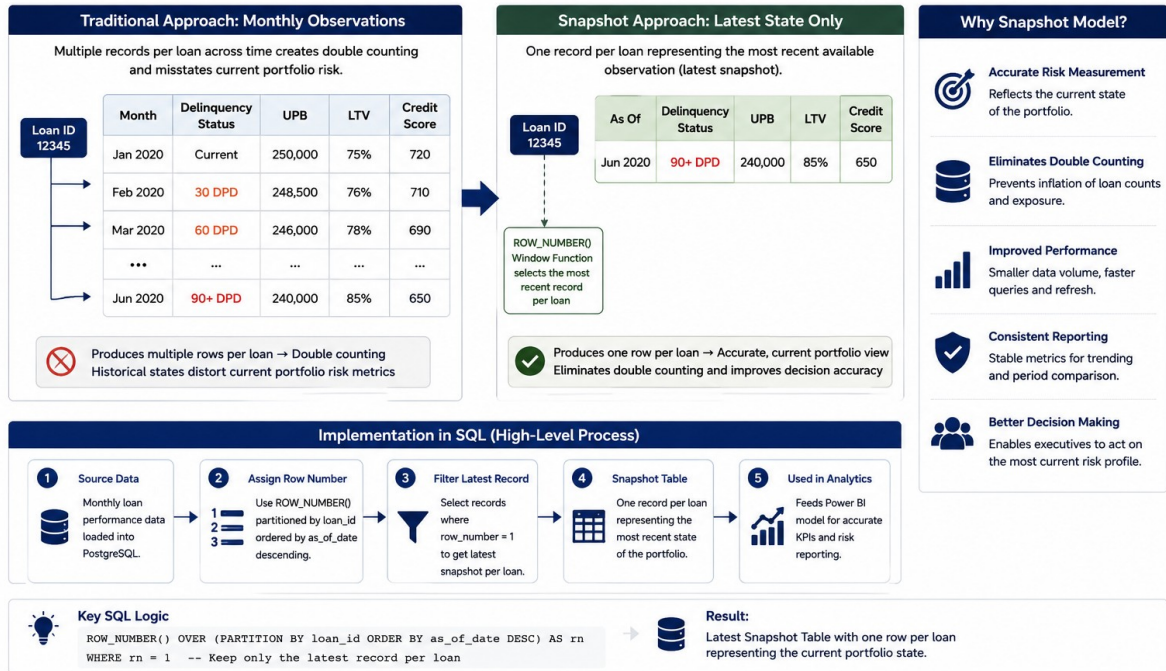


Figure 2. Snapshot Model Architecture

The grain problem and its solution: the traditional monthly observation model (left) generates multiple rows per loan causing double counting; the ROW_NUMBER() window function approach (right) selects exactly one current-state record per loan, producing accurate executive KPIs.

Side-by-side illustration: Left panel shows one loan appearing six times across monthly rows with accumulating UPB values. Right panel shows the same loan appearing once, retaining only its most recent observation. Highlights the double-counting problem and the snapshot solution.

The Latest-Snapshot Solution

The solution is to create a secondary analytical layer with a different grain: one row per unique loan, representing the most recent available observation for that loan. This is achieved by partitioning the panel by Loan Identifier and selecting the row with the maximum Monthly Reporting Period for each loan.

The resulting table — approximately 1.92 million unique loan records — has different characteristics from the panel. Total exposure is the current outstanding balance of each loan at its most recent observation: \$194 billion. Delinquency rate is the proportion of unique loans that are currently delinquent, not the proportion of monthly observations that recorded delinquency at any point: 0.65%. Current performing rate is the proportion of loans whose most recent status is current: 99.3%. These figures answer the actual management question — what does the portfolio look like right now?

TECHNICAL NOTE The snapshot approach is standard practice in institutional portfolio reporting. Banks' risk dashboards, regulatory reports, and board-level credit summaries all report current outstanding positions rather than cumulative historical balances. The snapshot model aligns the dashboard's outputs with the conventions that credit risk executives actually use.

Advantages of the Snapshot Model

- Eliminates double-counting of individual loans across reporting periods.

- Produces current-state KPIs that match executive expectations and industry reporting conventions.
- Reduces Power BI model size by approximately 98% relative to the full panel — from 84 million to 1.92 million rows — improving dashboard performance and refresh speed.
- Enables accurate cross-filtering and drill-down without risk of observation-level double-counting corrupting visual calculations.
- Consistent with how Fannie Mae's own portfolio surveillance reports represent current pool positions.

Trade-offs and Limitations

- Transition analysis is not possible from the snapshot alone. Roll rates (the probability of a loan moving from 30 DPD to 60 DPD in the next period) require the full panel with month-over-month comparisons.
- The snapshot captures a loan's most recent status — which may be current even if that loan was delinquent two months ago. Historically stressed but currently cured loans appear identical to loans that have never been delinquent.
- Vintage analysis requires joining the snapshot to origination date attributes, which are available in the snapshot but require careful interpretation because the snapshot represents each loan's current position, not its position at origination.

Coexistence of Both Models

The two models serve complementary purposes and neither can substitute for the other. The 84.18 million row panel was used for all SQL EDA, KPI development, and feature engineering work — where the full historical record provides the richest analytical signal. The 1.92 million row snapshot is the Power BI source — where current-state accuracy and performance are the governing requirements. Both the Executive Report and the Business Case Study explicitly document this architectural distinction and explain which figures originate from which model.

Interview guidance: *If asked why the dashboard shows \$194bn in exposure while the SQL analysis produced \$19.97 trillion, the correct answer is not 'the dashboard is filtered' or 'the numbers are different'. The correct answer is: 'The two figures come from different data grains. The SQL figure is the cumulative sum of monthly outstanding balances across all observation periods. The dashboard figure is the current outstanding balance of 1.92 million unique loans in the latest-snapshot model. Both figures are correct in their respective analytical contexts.' This distinction demonstrates genuine data modelling understanding.*

4 DASHBOARD DESIGN PHILOSOPHY

Executive-First Reporting

Every design decision in the dashboard was evaluated against one test: can a Chief Risk Officer extract the key insight within three seconds of looking at this visual or page? This is not a vague aspiration — it is an operational constraint that drove specific decisions about chart type, label placement, colour usage, and information density.

Executive-first design is different from analyst-first design. An analyst dashboard maximises information density — more metrics, more dimensions, more granularity per page. An executive dashboard maximises signal clarity — one primary message per page, supporting evidence visible but secondary, and a clear action implication at the bottom. This project was deliberately designed as the latter.

Progressive Narrative Sequencing

The six pages follow a deliberate analytical progression: each page answers one question and creates the frame for the next. The sequencing was not arbitrary — it mirrors the way credit risk analysis is actually presented in committee settings: start with portfolio health, then identify who is causing risk, then examine the structural context, then diagnose the geographic distribution, then explain the behavioural signals, and finally translate everything into an operational monitoring framework.

Figure 3. Dashboard Storytelling Framework



Figure 3. Dashboard Storytelling Framework

The six-page dashboard narrative framework — Strategic Flow: Overview → Diagnosis → Deep Dive → Context → Behaviour → Monitoring — with key KPIs, analytical purpose, and narrative flow for each page, governed by five design principles: Clarity, Consistency, Comparability, Actionability, and Timeliness.

Sequential diagram showing the business question each page answers and the connecting thread between consecutive pages. Illustrates the logical dependency structure of the narrative.

A key design constraint was that each page must add genuinely new analytical value — no page should repeat information already provided. This constraint was tested explicitly during design: if a visual could have appeared on a previous page without losing analytical value, it was removed or repositioned. LTV analysis, for example, was explicitly excluded from Page 2 (Borrower Risk) and reserved for Page 3 (Property and Loan) because LTV is a loan structural characteristic, not a borrower characteristic.

Consistent KPI Strip

The decision to maintain the same six KPI cards across all six pages was debated and deliberately confirmed. The alternative — page-specific KPI strips that change to show metrics relevant to each page's topic — was rejected for three reasons. First, executives navigating across pages need a constant orientation anchor. Second, changing KPI cards on every page forces the executive to re-learn the metrics on each navigation, interrupting analytical flow. Third, the six portfolio-level KPIs (exposure, delinquency rate, serious delinquency rate, performing rate, delinquent exposure, total loans) are relevant context for every analytical dimension — not just the executive overview page.

Filter Strategy

Each page exposes filters relevant to its analytical dimension: Page 2 includes Credit Score Band, Risk Segment, and Property State slicers. Page 3 includes Loan Purpose, Property State, and Risk Segment. Page 4 includes Property State and Risk Segment. All pages share consistent filter placement in the left sidebar panel. Filters were designed to allow executives to test specific hypotheses — 'what does the High Risk segment look like in California?' — without requiring navigation away from the current page.

DESIGN DECISION *The REMOVEFILTERS function was applied to KPI strip measures to ensure that applying a page-level filter (e.g., filtering to California) does not alter the portfolio-level KPI cards. The KPI strip always shows whole-portfolio figures, while the analytical visuals below reflect the filtered view. This design pattern prevents a common confusion: executives filtering to a state and then misreading the KPI cards as state-level metrics.*

Visual Type Hierarchy

Chart types were selected using a strict decision hierarchy based on the analytical question, not visual preference. The hierarchy applied throughout:

- Ranked comparison across categories → Horizontal bar chart (ordered by value).
- Distribution of a continuous variable → Vertical bar chart (ordered by natural sequence, e.g. risk bands).
- Two correlated dimensions simultaneously → Combo chart (bars for count/volume, line for rate).
- Geographic spatial pattern → Choropleth map (colouring by the diagnostic metric being tested, not by the most obvious metric).
- Multi-dimensional profile comparison → Conditional-formatted matrix (the only format that communicates multiple metrics across multiple entities simultaneously).
- Operational monitoring status → Structured cards with threshold and RAG indicator.

No pie charts, radar charts, or decorative visuals were used. Each rejected visual type was evaluated and specifically excluded for a documented reason — not avoided by default.

Colour Strategy

The colour palette — navy (#0D2B4E), teal (#1A6B7C), and gold (#C9A84C) — was selected to align with Fannie Mae's institutional brand and the visual conventions of financial services reporting. These colours communicate institutional credibility rather than creative expression. Within the dashboard, colour carries semantic meaning: gold accent rules and threshold lines signal the demarcation between analytical content

and structural framing. Navy backgrounds signal primary information. The consistent application of this palette across all six pages creates visual coherence that professional reviewers will recognise as intentional.

Page 6 — The Design Mode Change

Page 6 deliberately breaks from the design pattern of Pages 1–5 in one specific way: the KPI strip is replaced with five Active Monitoring Priority Cards. This is not an inconsistency — it is a intentional signal that the executive has moved from analysis mode to action mode. Pages 1–5 answer analytical questions about the portfolio. Page 6 answers the operational question: what should management actually do? The monitoring cards, with their explicit threshold values and RAG status indicators, communicate operational status rather than analytical insight. This mode change mirrors how real-world risk committee reporting transitions from portfolio analysis sections to monitoring and action sections.

5 DASHBOARD WALKTHROUGH — TECHNICAL REFERENCE

This section provides a technical reference for each dashboard page and every visual within it. The focus is on the reasoning behind each analytical and design choice — not on describing what is visible. For business interpretation of findings, refer to the Business Case Study.

Page 1 — Executive Portfolio Overview

Executive Business Question: How healthy is the overall mortgage portfolio today?

Business Objective: Provide a complete portfolio health orientation in a single page before any drill-down analysis.

Page 1 is intentionally structured as an orientation layer, not an analytical layer. Every visual is chosen for speed of comprehension over analytical depth. A CRO who has no time to navigate beyond Page 1 should still be able to answer all six headline portfolio questions from the KPI strip alone.

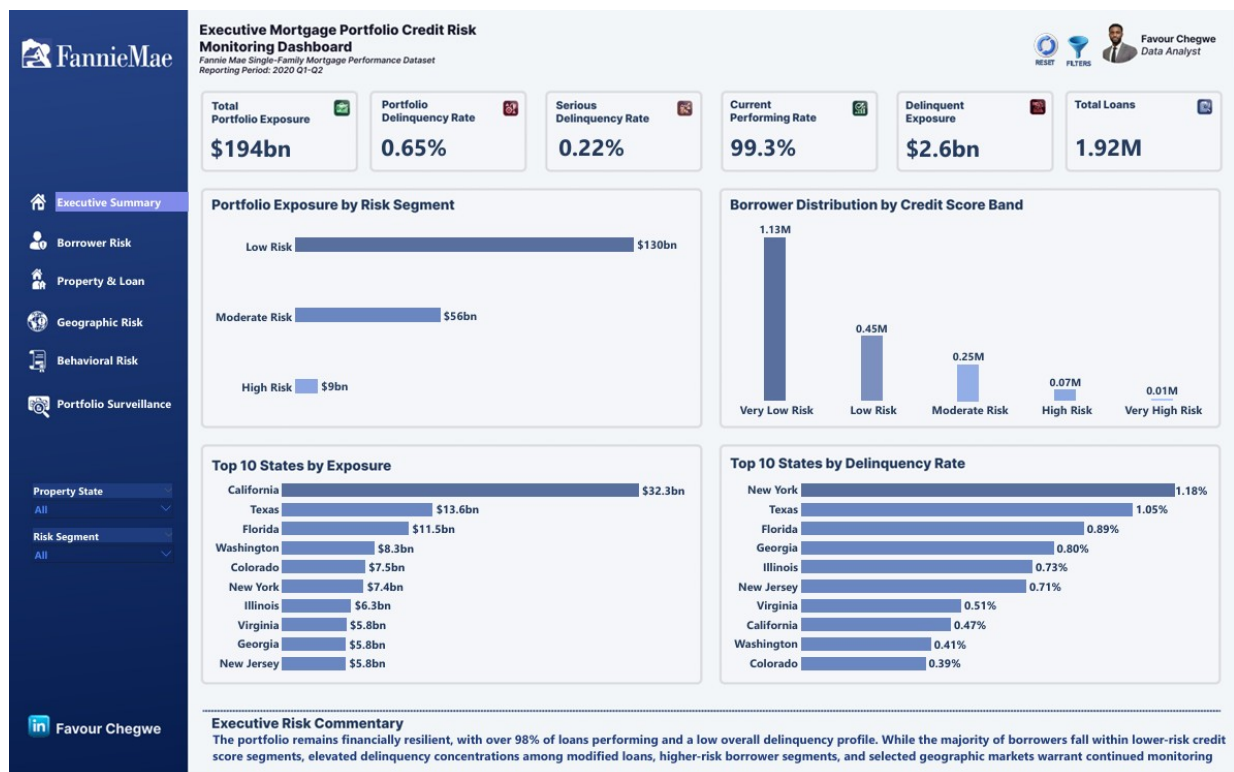


Figure 4. Page 1 — Executive Portfolio Overview

The portfolio health orientation page: six KPI cards (Portfolio Exposure \$194bn, Delinquency Rate 0.65%, Serious Delinquency 0.22%, Performing Rate 99.3%, Delinquent Exposure \$2.6bn, Total Loans 1.92M), supported by risk segment exposure, credit score distribution, and dual geographic charts — establishing that exposure concentration and credit quality risk are different phenomena.

Full-page screenshot showing the KPI strip, Portfolio Exposure by Risk Segment bar chart, Borrower Distribution by Credit Score Band vertical bar, Top 10 States by Exposure horizontal bar, and Top 10 States by Delinquency Rate horizontal bar.

◆ KPI Strip — Six Portfolio Cards

Business Question Answered: What are the six headline portfolio metrics at current snapshot date?

Visual Selected: Six equally-sized card visuals displaying scalar values with descriptive labels.

Alternatives Considered & Rejected: Page-specific KPI strip (rejected: executives need consistent orientation across all pages). Single summary card (rejected: insufficient information density for executive orientation).

Data Used: Snapshot fact table. All six measures use REMOVEFILTERS to ensure whole-portfolio values regardless of active page filters.

Measure Logic: Each measure calculates against the full snapshot table using CALCULATE with REMOVEFILTERS([Dimension]) to prevent slicer context from affecting headline numbers. Delinquency Rate = $\text{DIVIDE}(\text{COUNTROWS}(\text{FILTER}(\text{Snapshot}, [\text{delinquency_status}] <> "00")), \text{COUNTROWS}(\text{Snapshot}))$.

Key Insight: Modified Loans are the only segment currently breaching threshold at 22.8%. All other KPIs indicate a broadly healthy portfolio.

Business Interpretation: The KPI strip is the dashboard's executive anchor — it provides the absolute reference against which all segment-level findings on subsequent pages should be interpreted.

Executive Decision Supported: Confirms overall portfolio stability. Directs attention to whether serious delinquency (0.22%) is approaching the 0.5% surveillance threshold.

◆ Portfolio Exposure by Risk Segment

Business Question Answered: How is portfolio financial weight distributed across the three risk tiers?

Visual Selected: Horizontal bar chart with three bars (Low \$130bn, Moderate \$56bn, High \$9bn). Ordered by risk level ascending.

Alternatives Considered & Rejected: Donut chart (rejected: absolute dollar values are more management-relevant than percentage proportions; donut charts require mental conversion). Stacked bar (rejected: single dimension does not require stacking).

Data Used: Snapshot fact table. Risk Segment derived field and sum of Current UPB.

Measure Logic: $\text{SUM}([\text{current_upb}])$ grouped by $[\text{risk_segment}]$. Three segments only — clean and fast to interpret.

Key Insight: High Risk exposure is \$9bn — small relative to total. Low Risk holds \$130bn — 67% of total exposure.

Business Interpretation: The portfolio's structural concentration in Low Risk is a credit quality strength. The High Risk segment's small dollar size contextualises its elevated delinquency rate: high intensity, contained scale.

Executive Decision Supported: Calibrates the proportional response to High Risk delinquency. A segment holding 4.6% of exposure does not warrant portfolio-wide policy changes.

◆ Borrower Distribution by Credit Score Band

Business Question Answered: How is the population of loans distributed across credit quality tiers?

Visual Selected: Vertical bar chart with five bands ordered from Very Low Risk (left) to Very High Risk (right). Data labels show loan counts.

Alternatives Considered & Rejected: Histogram (rejected: bands are categorical, not continuous; bar chart is more precise for discrete categories). Line chart (rejected: there is no continuous trend to communicate).

Data Used: Snapshot fact table. Credit Score Band derived field and COUNTROWS per band.

Measure Logic: $\text{COUNTROWS}(\text{FILTER}(\text{Snapshot}, [\text{credit_score_band}] = "Very\ Low\ Risk"))$ replicated for each band. Simple count with no ratio — the absolute volumes communicate the distribution story.

Key Insight: 1.13M loans in Very Low Risk — 59% of the portfolio. Only 10K in Very High Risk.

Business Interpretation: The steeply declining volume from left to right is the reassurance visual. Even if Very High Risk delinquency rates are extreme, 10,000 loans out of 1.92 million is a manageable population.

Executive Decision Supported: Sets expectations for the delinquency-by-band analysis on Page 2. The distribution context prevents misinterpretation of high percentage delinquency rates as representing large absolute loan volumes.

◆ Top 10 States by Portfolio Exposure

Business Question Answered: Where is the portfolio's financial weight geographically concentrated?

Visual Selected: Horizontal bar chart, ranked by exposure descending. Labels show dollar values.

Alternatives Considered & Rejected: Choropleth map (rejected: state size disparities make small states nearly invisible; large states dominate visually regardless of risk. The bar chart enables precise magnitude comparison). Treemap (rejected: proportional area is harder to compare precisely than bar length).

Data Used: Snapshot fact table. Property State and SUM of Current UPB, limited to Top 10 by exposure.

Measure Logic: TOPN(10, SUMMARIZE(Snapshot, [property_state], "Exposure", SUM([current_upb])), [Exposure], DESC). Standard TOPN measure filtered to state dimension.

Key Insight: California at \$32.3bn — 16.6% of total exposure. Top three states represent ~29% of portfolio.

Business Interpretation: Exposure concentration of this magnitude creates systematic vulnerability to regional economic shocks. This visual plants the geographic risk question that Page 4 answers.

Executive Decision Supported: Prompts geographic stress scenario planning for high-concentration states, particularly California.

◆ Top 10 States by Delinquency Rate

Business Question Answered: Which geographic markets exhibit the highest current delinquency rates?

Visual Selected: Horizontal bar chart, ranked by delinquency rate descending.

Alternatives Considered & Rejected: Map overlay (rejected: delinquency rate comparison across a small number of named markets is clearer in a ranked bar than in geographic colour encoding).

Data Used: Snapshot fact table. Property State, Delinquency Flag, and COUNTROWS.

Measure Logic: DIVIDE(COUNTROWS(FILTER(SUMMARIZE context, [dq_flag] = 1)), COUNTROWS(...)) per state, ranked by rate.

Key Insight: New York leads at 1.18%. Critically, California — the largest exposure state — appears near the bottom of this chart at 0.47%.

Business Interpretation: The deliberate side-by-side placement of the exposure and delinquency charts is the most important design decision on Page 1. Their contrast communicates immediately that exposure concentration and credit quality risk are different problems affecting different states.

Executive Decision Supported: Identifies the geographic markets that warrant targeted monitoring on Page 4.

Page 2 — Borrower Risk Analysis

Executive Business Question: Which borrowers are most likely to become delinquent, and where is delinquent exposure financially concentrated?

Page 2 shifts from portfolio overview to borrower-level diagnosis. The page was designed around a specific tension: the segment with the highest delinquency rate (Very High Risk) is not the segment with the most delinquent exposure (Moderate Risk). Every visual on the page contributes to surfacing and explaining that tension.

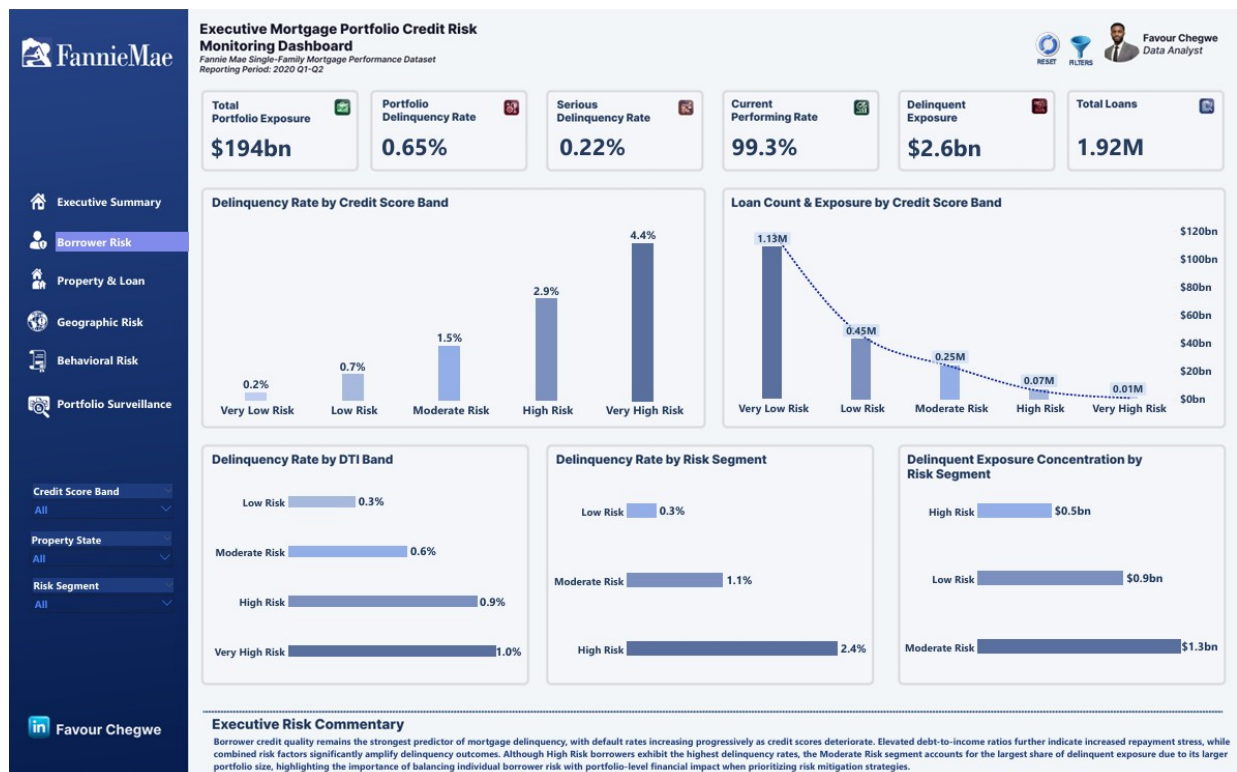


Figure 5. Page 2 — Borrower Risk Analysis

Five analytical visuals establishing that credit score is the dominant risk driver (22x delinquency differential), DTI is a secondary indicator, risk layering produces non-linear amplification, and the Moderate Risk segment carries the largest delinquent exposure (\$1.3bn) despite not having the highest delinquency rate.

Full-page screenshot showing all five visuals: Delinquency Rate by Credit Score Band, Loan Count & Exposure by Credit Score Band, Delinquency Rate by DTI Band, Delinquency Rate by Risk Segment, Delinquent Exposure Concentration by Risk Segment.

◆ Delinquency Rate by Credit Score Band

Business Question Answered: How steeply does delinquency risk escalate as credit quality deteriorates?

Visual Selected: Vertical bar chart ordered Very Low Risk to Very High Risk (left to right — ascending risk). Data labels show rates.

Alternatives Considered & Rejected: Line chart (rejected: implies continuous change between categories that are discrete bands; bar chart is more precise for categorical comparison). Horizontal bar (rejected: the ascending left-to-right risk progression reads more naturally in a vertical format).

Data Used: Snapshot fact table. Credit Score Band and Delinquency Rate measure.

Measure Logic: $\text{DIVIDE}(\text{COUNTROWS}(\text{FILTER}(\text{Snapshot}, [\text{dq_flag}]=1 \ \&\& \ [\text{credit_score_band}] = \text{"Very High Risk"})), \text{COUNTROWS}(\text{FILTER}(\text{Snapshot}, [\text{credit_score_band}] = \text{"Very High Risk"})))$ — replicated per band.

Key Insight: 22-fold difference between Very Low Risk (0.2%) and Very High Risk (4.4%). The gradient is steep and consistent.

Business Interpretation: Credit score is the single strongest underwriting variable. The visual gradient communicates this instantly without requiring the executive to process the numbers.

Executive Decision Supported: Empirical support for credit score-based approval cutoffs and tiered risk pricing.

◆ Loan Count and Exposure by Credit Score Band

Business Question Answered: How does portfolio volume and financial weight align with the delinquency risk gradient?

Visual Selected: Combo chart — vertical bars for loan count per band, dotted line for exposure. Shared X-axis (credit score band).

Alternatives Considered & Rejected: Two separate charts (rejected: showing both dimensions simultaneously on one visual is more powerful than requiring the executive to compare two charts mentally). Scatter plot (rejected: this is not a correlation between two continuous variables; it is a profile comparison across discrete bands).

Data Used: Snapshot fact table. Credit Score Band, COUNTROWS (bars), and SUM([current_upb]) (line).

Measure Logic: Bars: COUNTROWS per credit_score_band. Line: SUM([current_upb]) per credit_score_band on secondary axis. Both measures share the credit score band axis.

Key Insight: Both volume and exposure decline steeply from Very Low Risk to Very High Risk — confirming that the portfolio's financial weight is concentrated in the lower-risk bands.

Business Interpretation: This chart answers the implied question after every delinquency rate chart: 'But how many loans are actually in the risky segment?' The visual confirmation that volume and exposure are low in the high-risk bands prevents overreaction to high percentage delinquency rates.

Executive Decision Supported: Calibrates the financial materiality of high-risk segment delinquency rates against their portfolio weight.

◆ Delinquency Rate by DTI Band

Business Question Answered: Does debt-to-income ratio independently predict delinquency outcomes?

Visual Selected: Horizontal bar chart ordered Low Risk to Very High Risk (top to bottom — ascending risk for horizontal layout).

Alternatives Considered & Rejected: Combo with credit score (rejected: combining two independent variables in one visual creates interpretive confusion. Each risk driver deserves its own visual to demonstrate independent contribution).

Data Used: Snapshot fact table. DTI Band derived field and Delinquency Rate measure.

Measure Logic: Same DIVIDE logic as credit score band, grouped by dti_band.

Key Insight: 0.3% to 1.0% range — consistent direction but moderate differentiation. DTI is a secondary, not primary, risk driver.

Business Interpretation: DTI confirms the directional relationship between payment capacity and delinquency. The modest differentiation range correctly positions DTI as a supporting variable in the risk-layering framework rather than a primary underwriting threshold.

Executive Decision Supported: Justifies including DTI in the risk-layering composite without overweighting it in single-variable analysis.

◆ Delinquency Rate by Risk Segment

Business Question Answered: What is the delinquency outcome when multiple risk factors are present simultaneously?

Visual Selected: Horizontal bar chart with three segments (Low, Moderate, High). Ordered by risk ascending.

Alternatives Considered & Rejected: Grouped bar alongside individual factor charts (rejected: the risk segment visual should stand alone to emphasise that layered risk is a distinct concept from individual factor risk. Combining them would blur the analytical distinction).

Data Used: Snapshot fact table. Risk Segment derived field and Delinquency Rate measure.

Measure Logic: Risk Segment is a pre-computed column in the snapshot table derived from Risk Factor Count (0=Low, 1=Moderate, 2-3=High). Delinquency rate calculated as DIVIDE(delinquent count, total count) per segment.

Key Insight: High Risk at 2.4% — eight times Low Risk at 0.3%. The jump from Moderate to High is proportionally larger than from Low to Moderate.

Business Interpretation: Risk-layering produces compounding rather than additive risk amplification. This finding cannot be derived from individual credit score, LTV, or DTI analysis alone — it is an emergent property of factor combination.

Executive Decision Supported: Empirical foundation for risk-layering underwriting policy requiring enhanced compensating factors when two or more risk conditions are present.

◆ Delinquent Exposure Concentration by Risk Segment

Business Question Answered: Which risk segment holds the largest share of financially delinquent exposure?

Visual Selected: Horizontal bar chart showing dollar delinquent exposure per segment. Ordered by exposure value.

Alternatives Considered & Rejected: Donut chart (rejected: absolute dollar values are more management-relevant than proportional shares. The actual delinquent dollar amount is the financially meaningful metric). Stacked bar with risk segment (rejected: cleaner as a standalone to allow the Moderate Risk finding to stand out without visual competition).

Data Used: Snapshot fact table. Risk Segment, Delinquency Flag, and SUM of Current UPB.

Measure Logic: SUM([current_upb]) filtered to delinquent records, grouped by risk_segment.

Key Insight: Moderate Risk holds \$1.3bn — the largest delinquent exposure — despite not having the highest delinquency rate.

Business Interpretation: This is the page's most analytically sophisticated finding. Delinquency rate and delinquent exposure tell different stories. The Moderate Risk segment's large volume means it accounts for more total delinquent dollars than the smaller but more intensely delinquent High Risk segment.

Executive Decision Supported: Mandates exposure-weighted monitoring in addition to delinquency-rate monitoring. Pure rate-based monitoring would misallocate surveillance resources away from Moderate Risk.

Page 3 — Property and Loan Characteristics

Executive Business Question: Which loan and property structures carry elevated delinquency risk, and do they represent a material portfolio concentration?

Page 3 shifts the analytical lens from borrower characteristics (Page 2) to structural and collateral characteristics. The key design principle: every visual on this page must add information that cannot be derived from the borrower risk analysis. LTV, loan purpose, and property type are structural variables fixed at origination — they capture risk dimensions that credit score and DTI do not explain.

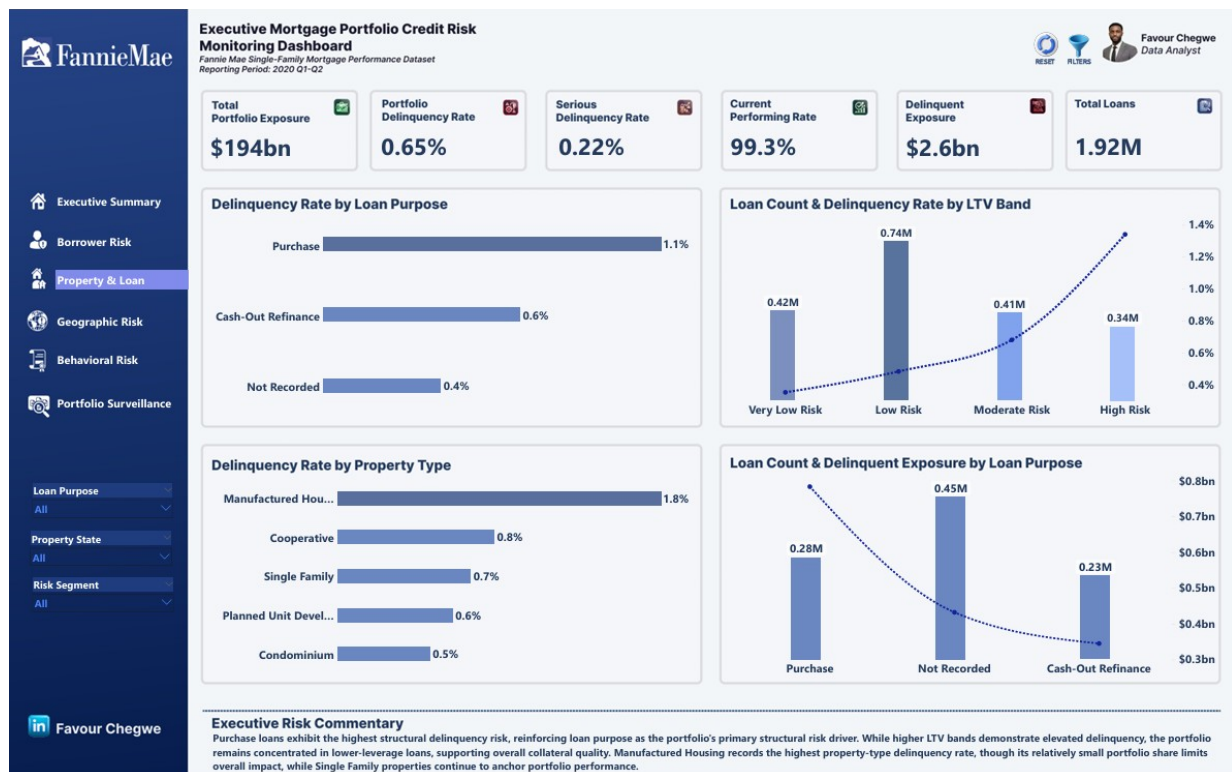


Figure 6. Page 3 — Property and Loan Characteristics

Four structural risk visuals: loan purpose delinquency (purchase 1.1% vs refinance 0.6%), LTV band risk gradient with portfolio concentration profile, property type performance (Manufactured Housing 1.8x), and loan purpose delinquent exposure — confirming purchase loans as the primary structural risk driver.

Full-page screenshot showing Delinquency Rate by Loan Purpose, Loan Count & Delinquency Rate by LTV Band combo, Delinquency Rate by Property Type, and Loan Count & Delinquent Exposure by Loan Purpose combo.

◆ Delinquency Rate by Loan Purpose

Business Question Answered: Do loans originated for different purposes exhibit materially different delinquency outcomes?

Visual Selected: Horizontal bar chart ordered by delinquency rate descending.

Alternatives Considered & Rejected: Pie chart (rejected: three categories at different absolute sizes is better communicated by bar than pie). Grouped with property type (rejected: loan purpose and property type are separate structural dimensions requiring separate analytical treatment).

Data Used: Snapshot fact table. Loan Purpose field (full description labels applied during data standardisation) and Delinquency Rate measure.

Measure Logic: $\text{DIVIDE}(\text{delinquent count, total count})$ per loan_purpose. 'Not Recorded' category retained as a separate bar rather than excluded.

Key Insight: Purchase loans at 1.1% — approximately double Cash-Out Refinance at 0.6%.

Business Interpretation: The purchase loan differential is real but contextualised by the 2020 Q1-Q2 reporting period. Purchase originations during this window were immediately exposed to COVID-19 income disruption during their most financially vulnerable early months. Vintage adjustment across subsequent cohorts is required to confirm structural versus timing attribution.

Executive Decision Supported: Justifies tracking purchase loans separately from refinance loans. Supports vintage-adjusted monitoring recommendation.

◆ Loan Count and Delinquency Rate by LTV Band

Business Question Answered: How does borrower leverage position affect delinquency risk, and where is the portfolio concentrated by LTV?

Visual Selected: Combo chart — vertical bars for loan count per LTV band, dotted line for delinquency rate.

Alternatives Considered & Rejected: Scatter plot LTV vs DQ rate at loan level (rejected: too granular for executive page; segment-level aggregation is more actionable). Bar chart for rate only (rejected: showing only rate without volume misses the portfolio concentration story).

Data Used: Snapshot fact table. LTV Band derived field, COUNTROWS (bars), Delinquency Rate measure (line).

Measure Logic: Bars: COUNTROWS per ltv_band. Line: DIVIDE(delinquent count, total count) per ltv_band on secondary axis.

Key Insight: Delinquency rises from 0.4% (Very Low Risk LTV) to 1.4% (High Risk LTV). Portfolio is concentrated in the Low LTV band (0.74M loans).

Business Interpretation: The portfolio's concentration in lower-LTV bands reduces aggregate loss severity because equity cushion limits loss given default. The LTV band analysis is the collateral quality counterpart to the credit quality analysis on Page 2.

Executive Decision Supported: Supports maintaining LTV discipline in origination as the primary collateral quality lever.

◆ Delinquency Rate by Property Type

Business Question Answered: Does collateral type exhibit meaningful delinquency variation across the portfolio?

Visual Selected: Horizontal bar chart ordered by delinquency rate descending.

Alternatives Considered & Rejected: Table/matrix (considered: with five categories, a table would work but the bar chart communicates magnitude differences more immediately at executive scale). Map by property type (rejected: not a spatial relationship).

Data Used: Snapshot fact table. Property Type field (full description labels) and Delinquency Rate measure.

Measure Logic: DIVIDE(delinquent count, total count) per property_type. Five categories.

Key Insight: Manufactured Housing at 1.8% — 3.6 times Condominium at 0.5%. Single Family at 0.7% near portfolio average.

Business Interpretation: Property type differentiation is meaningful (3.6x spread) but secondary to borrower risk. The chart was originally considered for replacement with a matrix following EDA showing weak differentiation — the post-snapshot rebuild confirmed sufficient variation to warrant a dedicated visual.

Executive Decision Supported: Identifies Manufactured Housing as a segment requiring enhanced underwriting scrutiny despite its limited portfolio share.

◆ Loan Count and Delinquent Exposure by Loan Purpose

Business Question Answered: Does the purchase loan delinquency finding translate into proportionally higher delinquent exposure?

Visual Selected: Combo chart — vertical bars for loan count per loan purpose, dotted line for delinquent exposure. Loan count on primary axis, delinquent exposure on secondary.

Alternatives Considered & Rejected: Two separate charts (rejected: the combination of volume and exposure on one visual directly connects the rate finding from Chart 1 to its financial consequence. Separation would require mental connection across two charts).

Data Used: Snapshot fact table. Loan Purpose, COUNTROWS (bars), and SUM([current_upb]) filtered to delinquent records (line).

Measure Logic: Bars: COUNTROWS per loan_purpose. Line: SUM([current_upb]) WHERE [dq_flag]=1 per loan_purpose.

Key Insight: Purchase loans: highest count (0.28M) and highest delinquent exposure (\$0.7bn). Ordering of rate, count, and delinquent exposure all converge on purchase loans.

Business Interpretation: The convergence of three independent metrics — delinquency rate (Chart 1), loan count, and delinquent exposure (this chart) — on purchase loans as the highest-risk purpose creates a robust, multi-dimensional finding that is harder to dismiss than a single-metric result.

Executive Decision Supported: Complete structural risk case for purchase loan separation in portfolio monitoring and the vintage analysis recommendation.

Page 4 — Geographic Risk Analysis

Executive Business Question: Why are certain geographic markets riskier than others?

Page 4 is the dashboard's most analytically sophisticated page. The reframe from 'which states are risky' to 'why are certain states riskier' was the most significant design evolution across the entire project. The four visuals are sequenced as a diagnostic investigation: establish the geographic pattern (map) → test the borrower composition hypothesis (combo chart) → confirm with a second independent behavioural signal (modification bar) → synthesise across all dimensions (matrix).

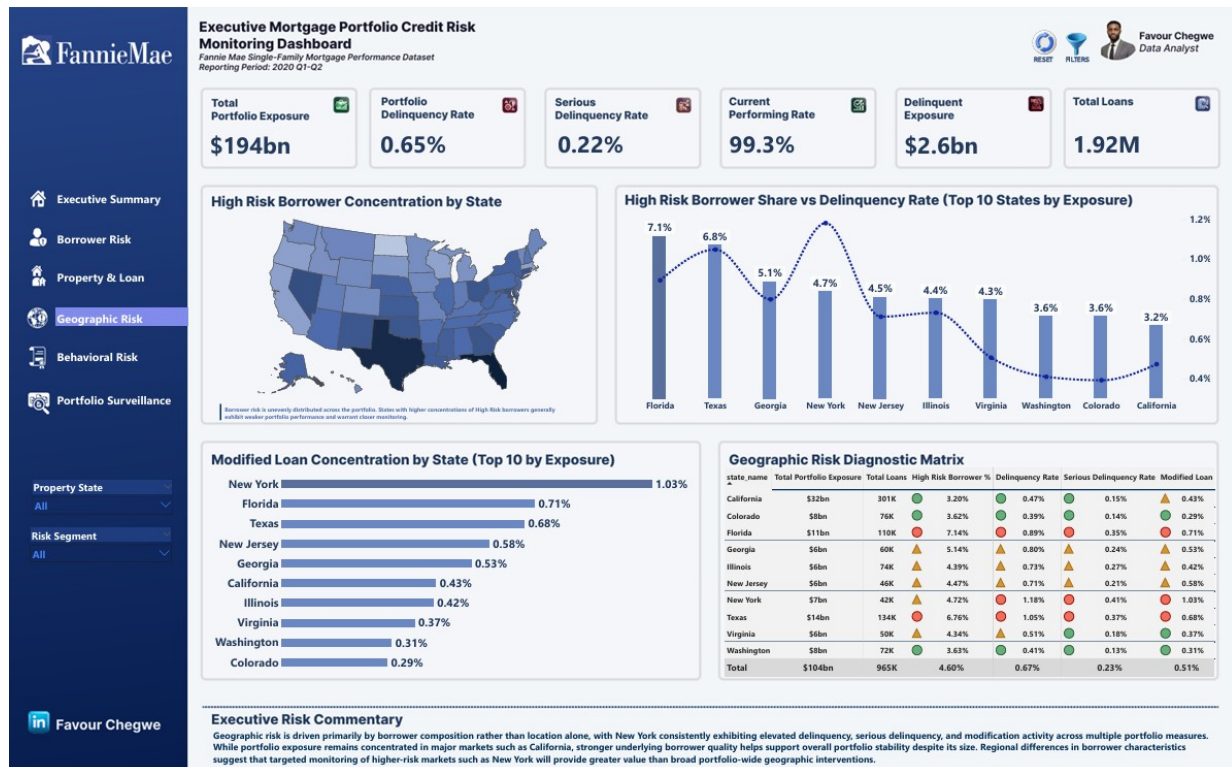


Figure 7. Page 4 — Geographic Risk Analysis

Four-visual diagnostic investigation: High Risk borrower concentration choropleth map (entry point), High Risk share vs delinquency rate combo chart (hypothesis test), modified loan concentration ranking (second independent signal), and conditional-formatted diagnostic matrix (synthesis) — proving that geographic risk reflects borrower composition, not location.

Full-page screenshot showing the High Risk Borrower Concentration choropleth map, High Risk Borrower Share vs Delinquency Rate combo chart, Modified Loan Concentration horizontal bar, and Geographic Risk Diagnostic Matrix with conditional formatting.

◆ High Risk Borrower Concentration by State (Shape Map)

Business Question Answered: Is High Risk borrower concentration distributed geographically in a pattern that correlates with state-level delinquency?

Visual Selected: Choropleth map coloured by High Risk Borrower % per state. Darker shading = higher concentration.

Alternatives Considered & Rejected: Delinquency rate map (considered and rejected: Page 1 already shows delinquency by state rankings. The diagnostic question for Page 4 is whether borrower composition explains those

delinquency rates. Colouring the map by High Risk share directly tests that hypothesis). Bubble map (rejected: bubble size on a map is harder to read than colour intensity for this use case).

Data Used: Snapshot fact table aggregated to state level. High Risk Borrower % = COUNTROWS(High Risk loans) / COUNTROWS(all loans) per state.

Measure Logic: State-level aggregation measure: DIVIDE(COUNTROWS(FILTER(Snapshot,[risk_segment]="High Risk")), COUNTROWS(Snapshot)) per property_state.

Key Insight: Florida (7.14%) and Texas (6.76%) show high concentrations. California (3.20%) shows low concentration despite leading in exposure.

Business Interpretation: The map is the entry point to the diagnostic page. It establishes the geographic pattern of the variable being tested (borrower quality composition) before the next visual confirms whether that pattern actually explains delinquency outcomes.

Executive Decision Supported: Sets up the diagnostic question: do states with higher High Risk concentrations also have higher delinquency rates?

◆ High Risk Borrower Share vs Delinquency Rate — Top 10 States by Exposure

Business Question Answered: Does High Risk borrower concentration explain state-level delinquency rates for the portfolio's most financially significant markets?

Visual Selected: Combo chart — bars for High Risk Borrower % per state, line for Delinquency Rate per state. States ordered by portfolio exposure (California = highest exposure, first bar).

Alternatives Considered & Rejected: Scatter plot (considered: would directly show correlation between the two variables. Rejected: with only 10 states, scatter plot loses the context of which state is which without crowded labels. Combo chart with named X-axis preserves state identity). Two separate bar charts (rejected: comparison across two measures for the same states is cleaner in one combo visual).

Data Used: Snapshot fact table aggregated to state level. High Risk % (bars) and Delinquency Rate (line) per property_state, filtered to top 10 by exposure.

Measure Logic: Bars: DIVIDE(High Risk count, total count) per state. Line: DIVIDE(delinquent count, total count) per state. TOPN filter applied at the state dimension.

Key Insight: Florida (7.14% High Risk, 0.89% DQ) and Texas (6.76%, 1.05%) show both measures elevated. California (3.20%, 0.47%) shows both measures below average. The pattern confirms the hypothesis.

Business Interpretation: The diagnostic hypothesis is confirmed: geographic risk tracks borrower quality composition. States where we see above-average High Risk borrower concentrations also show above-average delinquency rates, with California as the critical counterexample that validates the model.

Executive Decision Supported: Geographic risk management should focus on the borrower quality dimension within each market, not just the state location. California's large exposure requires scenario monitoring, not credit quality escalation.

◆ Modified Loan Concentration by State — Top 10 by Exposure

Business Question Answered: Do high-delinquency states also exhibit elevated modification activity — providing a second independent confirmation of the risk profile?

Visual Selected: Horizontal bar chart ordered by modification rate descending. New York leads dramatically.

Alternatives Considered & Rejected: Line chart over time (rejected: this is a cross-sectional state comparison, not a time series). Integration with the combo chart (rejected: modification rate is a separate dimension from High Risk borrower share; it needs its own visual to communicate as an independent confirmation rather than an additional correlated metric).

Data Used: Snapshot fact table aggregated to state level. Modified Loan % = COUNTROWS(Modified loans) / COUNTROWS(all loans) per state.

Measure Logic: DIVIDE(COUNTROWS(FILTER(Snapshot,[modification_flag]="Y")), COUNTROWS(Snapshot)) per property_state, filtered to top 10 by exposure.

Key Insight: New York at 1.03% — more than double Florida (0.71%) in third place. California at 0.43% below average.

Business Interpretation: Two independent metrics — High Risk borrower share and modification concentration — both point to the same markets as elevated risk. The independence of these signals strengthens the diagnostic conclusion. If only one metric elevated New York, the finding would be less robust.

Executive Decision Supported: Confirms New York as the portfolio's highest-priority geographic monitoring market across multiple independent risk dimensions.

◆ Geographic Risk Diagnostic Matrix

Business Question Answered: What is the complete multi-dimensional risk profile of each of the 10 most financially significant geographic markets?

Visual Selected: Conditional-formatted matrix — 10 state rows × 6 metric columns (Exposure, Loans, High Risk %, DQ Rate, Serious DQ Rate, Modified %). RAG colouring applied per column based on portfolio-relative thresholds.

Alternatives Considered & Rejected: Heatmap visual (considered: a heatmap would provide the multi-dimensional colouring without the precise values. Rejected: precise values matter for executive committee discussion — the matrix provides both). Multiple charts (rejected: six individual charts for 10 states would be unreadable at dashboard scale). Table without conditional formatting (rejected: a plain table requires the reader to mentally compare values; conditional formatting communicates relative standing instantly).

Data Used: Snapshot fact table aggregated to state level. Six separate measures calculated per state: SUM UPB, COUNTROWS, High Risk %, DQ Rate, Serious DQ Rate, Modified Loan %.

Measure Logic: Each column uses its own aggregation measure applied in row context. Conditional formatting applied via DAX-driven background colour rules: values above portfolio average → red/amber; below → green.

Key Insight: New York is the only state elevated across all five risk metrics simultaneously. California shows green across almost all metrics despite its largest exposure.

Business Interpretation: The matrix is the executive synthesis of the entire geographic analysis. One table communicates what would otherwise require six separate charts and mental integration across all of them. The conditional formatting converts the data into an immediately readable risk profile.

Executive Decision Supported: Definitively separates the California exposure concentration problem (scenario stress testing required) from the New York credit quality problem (increased monitoring frequency required).

Page 5 — Behavioural Risk Analysis

Executive Business Question: How does modification status influence delinquency severity, and does this effect persist independently of borrower risk classification?

Page 5 was designed to answer one specific analytical question that cannot be inferred from Pages 1–4: is modification status an independent risk predictor, or does it simply identify the same loans that the risk segment classification already flags? The three-visual layout was deliberately minimal — the finding is powerful enough that visual complexity would dilute rather than strengthen it.

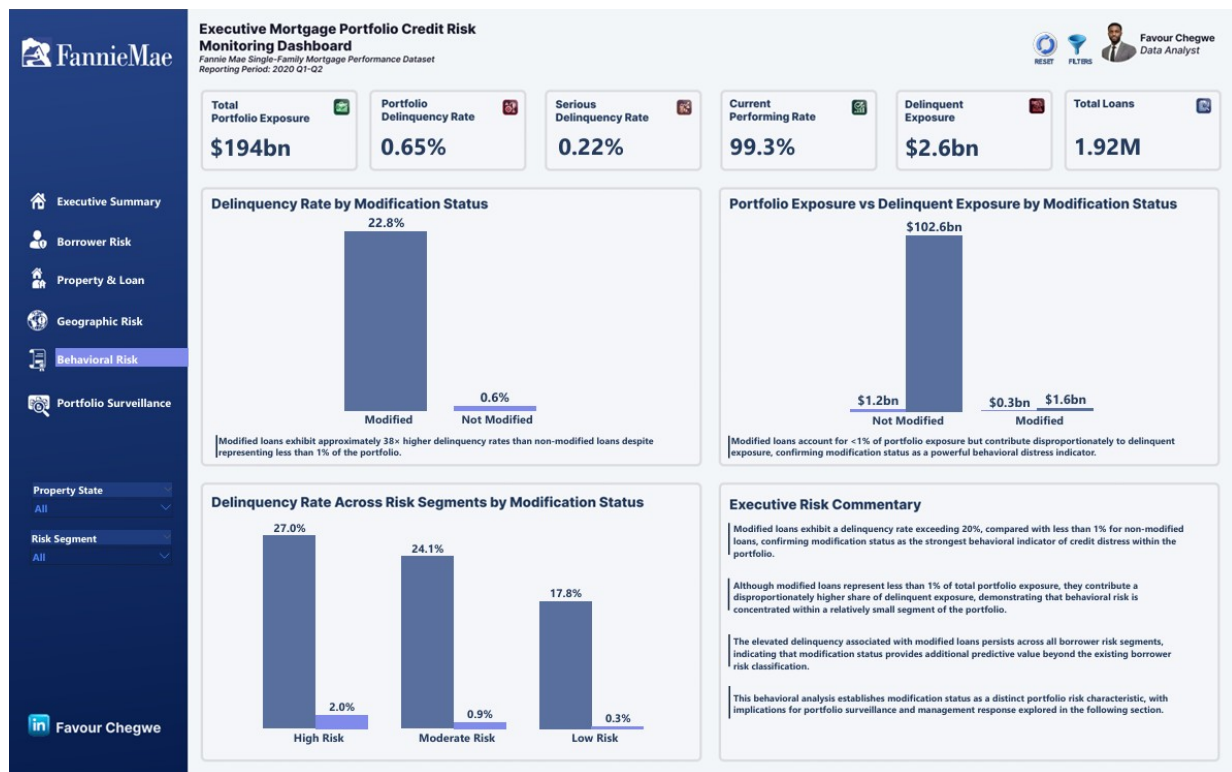


Figure 8. Page 5 — Behavioural Risk Analysis

Three-visual behavioural proof: 22.8% vs 0.6% modification delinquency comparison (impact), \$1.6bn vs \$0.3bn delinquent exposure disproportion (financial proof), and clustered cross-segment bars (independence proof) — confirming modification status as an independent risk signal that carries predictive value beyond existing risk segment classification.

Full-page screenshot showing the two-bar Delinquency Rate by Modification Status comparison, the four-bar Portfolio Exposure vs Delinquent Exposure comparison, and the clustered Delinquency Rate Across Risk Segments by Modification Status chart.

◆ Delinquency Rate by Modification Status

Business Question Answered: How dramatically does modification status affect a loan's likelihood of current delinquency?

Visual Selected: Two-bar comparison chart — Modified (22.8%) vs Not Modified (0.6%). Large data labels for immediate impact.

Alternatives Considered & Rejected: KPI cards (considered: two large comparison cards would also communicate the magnitude. Rejected: bar chart preserves the visual proportion between the two values — the height difference directly communicates the 38x multiple without requiring mental arithmetic). Gauge chart (rejected: gauges are appropriate for single-value metrics against a target, not for two-category comparison).

Data Used: Snapshot fact table. Modification Flag (Y/N standardised to 'Modified'/'Not Modified') and Delinquency Rate measure.

Measure Logic: $\text{DIVIDE}(\text{delinquent count}, \text{total count})$ per modification_flag value. Two categories only.

Key Insight: 22.8% for modified vs 0.6% for non-modified. A 38-fold differential.

Business Interpretation: The scale of this differential is designed to be visually arresting. The bar chart's proportional height difference makes the 38x multiple visible without the executive needing to calculate it. The caption reinforces the '38x' framing explicitly.

Executive Decision Supported: Confirms modification status as the single strongest behavioural delinquency predictor in the portfolio. Establishes that modified loans require monitoring distinct from the general portfolio.

◆ Portfolio Exposure vs Delinquent Exposure by Modification Status

Business Question Answered: Does the modification delinquency rate finding translate into a disproportionate concentration of delinquent financial exposure?

Visual Selected: Four-bar comparison — two pairs of bars (Not Modified: \$102.6bn total / \$1.2bn delinquent; Modified: \$0.3bn total / \$1.6bn delinquent).

Alternatives Considered & Rejected: Percentage bar showing share of delinquent exposure (rejected: absolute dollar values are more impactful for executive decision-making than proportional shares. The fact that \$1.6bn in delinquent exposure comes from a \$0.3bn total pool communicates the disproportionality more viscerally in absolute terms). Scatter plot (rejected: not appropriate for comparing category pairs at different scales).

Data Used: Snapshot fact table. Modification Flag, SUM([current_upb]) (total exposure), and SUM([current_upb]) filtered to delinquent loans (delinquent exposure).

Measure Logic: Total Exposure measure: SUM([current_upb]) per modification_flag. Delinquent Exposure measure: SUM([current_upb]) WHERE [dq_flag]=1 per modification_flag.

Key Insight: Modified loans: \$0.3bn total exposure, \$1.6bn delinquent exposure. More delinquent exposure than total exposure — possible because some modified loans have UPBs reported at removal or deferral-adjusted balances.

Business Interpretation: The financial proof of the concentration argument. Modified loans generate delinquent exposure multiple times larger than their total portfolio exposure would predict. This single chart justifies dedicating an entire monitoring programme to a segment representing less than 1% of the portfolio.

Executive Decision Supported: Provides financial quantification for Recommendation 1: dedicated modified loan monitoring delivers the highest risk management return per dollar of resource invested.

◆ Delinquency Rate Across Risk Segments by Modification Status

Business Question Answered: Does modification status add delinquency risk independently of the borrower's existing risk segment classification?

Visual Selected: Clustered bar chart — three risk segment groups on X-axis, two bars per group (Modified vs Not Modified). Six bars total.

Alternatives Considered & Rejected: Faceted line chart (rejected: the within-group comparison of two bars is more natural in a clustered bar than as connected lines across categories). Single-series bar per modification status (rejected: showing the two series side-by-side within each risk segment is the only way to visually prove the within-segment comparison).

Data Used: Snapshot fact table. Risk Segment, Modification Flag, and Delinquency Rate measure in a 3x2 matrix calculation.

Measure Logic: DIVIDE(delinquent count, total count) per [risk_segment] AND [modification_flag] combination. Six separate measures or a single measure that responds to both filter contexts.

Key Insight: Within High Risk: Modified 27.0% vs Not Modified 2.0%. Within Low Risk: Modified 17.8% vs Not Modified 0.3%. The modification premium exists and is large in every segment.

Business Interpretation: This is the independence proof chart. If modification were merely proxying for risk segment, modified and non-modified loans within the same segment would show similar rates. The 13x to 59x gaps within each segment prove that modification status carries risk information completely independent of the existing risk classification.

Executive Decision Supported: Justifies treating modification status as a standalone portfolio risk variable that requires its own monitoring infrastructure, separate from the risk segment framework.

Page 6 — Portfolio Surveillance

Executive Business Question: Which portfolio segments require continuous management attention, and which are currently in breach of risk thresholds?

Page 6 is architecturally different from Pages 1–5. The shift from analytical reporting to operational monitoring required deliberate design choices: the KPI strip is replaced, the chart types change from analytical to operational, and the page closes with text panels (Surveillance Framework and Executive Risk

Commentary) rather than additional visuals. The entire page design reflects the difference between 'here is what the analysis found' and 'here is what management needs to track.'

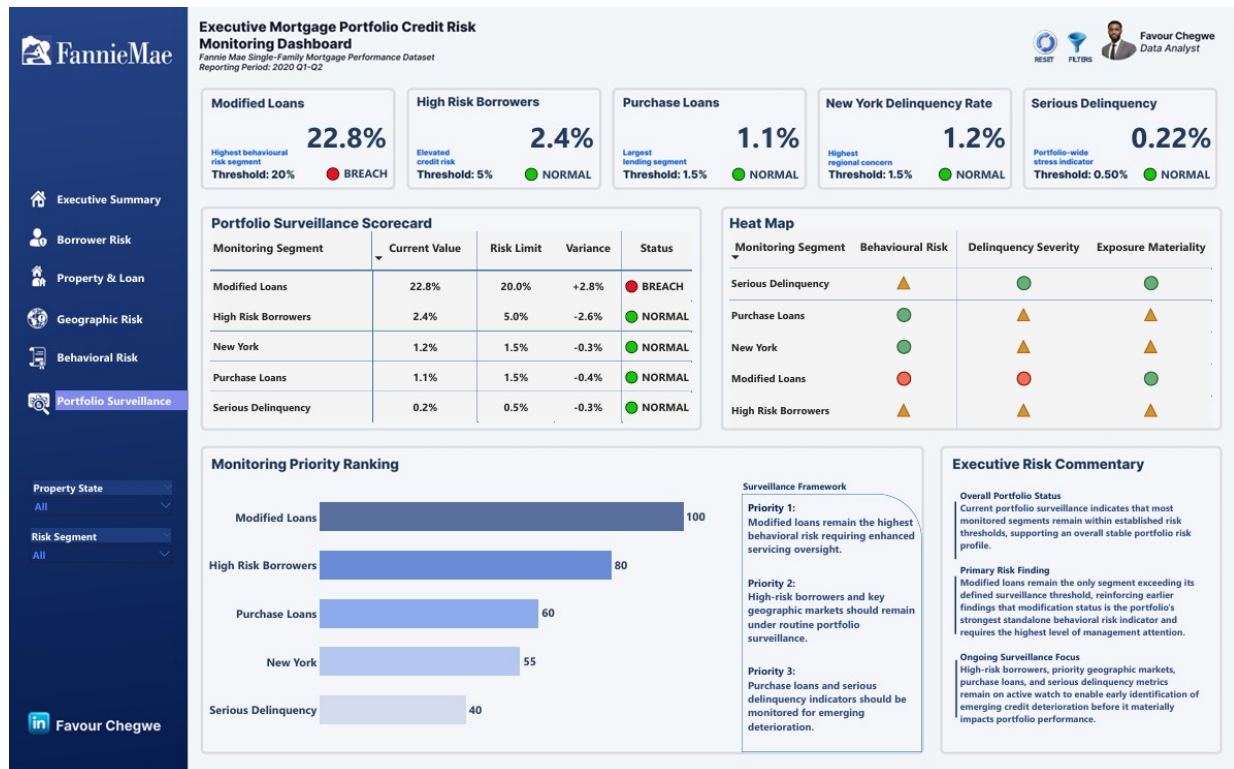


Figure 9. Page 6 — Portfolio Surveillance

Operational monitoring framework: five Active Monitoring Priority Cards (Modified Loans BREACH at 22.8% vs 20.0% threshold), Portfolio Surveillance Scorecard with variance tracking, Risk Heat Map across three dimensions, indexed Monitoring Priority Ranking, and Surveillance Framework text panel — demonstrating the mode change from analytical reporting to operational management.

Full-page screenshot showing the five Active Monitoring Priority Cards, Portfolio Surveillance Scorecard matrix, Risk Heat Map, Monitoring Priority Ranking bar chart, Surveillance Framework text panel, and Executive Risk Commentary.

◆ Active Monitoring Priority Cards (replacing KPI strip)

Business Question Answered: Which monitored segments are currently in breach of defined risk thresholds, and which are within normal limits?

Visual Selected: Five large KPI-style cards replacing the standard portfolio KPI strip. Each card: segment name, current value, threshold, RAG status indicator.

Alternatives Considered & Rejected: Standard KPI strip continued from Pages 1–5 (rejected: Page 6 is operational, not analytical. The mode change from analysis to surveillance is signalled by changing the top-of-page element. Continuing the portfolio KPI strip would blur this transition).

Data Used: Snapshot fact table. Delinquency rate measures for each of the five monitored segments, compared against hard-coded threshold values.

Measure Logic: Each card: $\text{DIVIDE}(\text{delinquent count}, \text{total count})$ per segment-specific filter. Threshold comparison: $\text{IF}([\text{segment_dq_rate}] > [\text{threshold_value}], \text{"BREACH"}, \text{"NORMAL"})$. RAG colour applied via conditional formatting on the status field.

Key Insight: Modified Loans: 22.8% vs 20.0% threshold — BREACH. All others: within limits.

Business Interpretation: At a single glance, the CRO knows there is one active breach and four normal segments. The card format mirrors institutional risk committee reporting conventions — status first, details in the scorecard below.

Executive Decision Supported: Confirms that only the Modified Loan segment requires immediate escalation. All others remain in monitoring mode.

◆ Portfolio Surveillance Scorecard

Business Question Answered: What is the precise current value, risk limit, variance, and status of every monitored segment?

Visual Selected: Conditional-formatted matrix — five rows (segments) × four columns (Current Value, Risk Limit, Variance, Status). Red fill for breach, green for normal.

Alternatives Considered & Rejected: Dashboard cards with embedded tables (rejected: the matrix format provides precision and comparability across segments simultaneously. Cards would require individual inspection of each segment). Bar chart of variances (rejected: the combination of current value, limit, and status requires a table structure that a bar chart cannot accommodate).

Data Used: Five segment-level delinquency rate measures, threshold constants, variance calculations.

Measure Logic: Variance = [current_value] - [threshold_value]. Positive = breach. Conditional formatting: background color = IF(Variance > 0, Red fill, Green fill).

Key Insight: Modified Loans: +2.8% variance (only positive variance). All other segments: negative variance (within limits).

Business Interpretation: The scorecard provides the precision layer behind the monitoring cards. It translates broad status indicators into specific numbers suitable for committee minutes and formal risk reporting.

Executive Decision Supported: Provides the specific figures needed for risk committee documentation and threshold review decisions.

◆ Risk Heat Map

Business Question Answered: Across three risk dimensions (Behavioural Risk, Delinquency Severity, Exposure Materiality), which segments show multi-dimensional elevation?

Visual Selected: Matrix of five monitoring segments × three risk dimensions with RAG indicators per cell.

Alternatives Considered & Rejected: Spider/radar chart (rejected: radar charts are difficult to read for absolute comparison across segments. The matrix provides the same multi-dimensional profile in a more precise, scannable format). Stacked bar by dimension (rejected: RAG status is categorical, not additive — stacking would misrepresent the nature of the information).

Data Used: Derived assessment of each segment across three dimensions based on portfolio-relative comparisons.

Measure Logic: Each cell: IF([segment_metric] > [portfolio_average_for_dimension], amber or red, green). Thresholds for each dimension applied independently.

Key Insight: Modified Loans: red across Behavioural Risk and Delinquency Severity. New York and Purchase Loans: amber in Delinquency Severity and Exposure Materiality.

Business Interpretation: The heat map reveals nuance that the scorecard cannot: while Modified Loans is the only formal threshold breach, New York and Purchase Loans carry multi-dimensional amber signals suggesting elevated future risk. The heat map enables pre-emptive monitoring before a threshold breach occurs.

Executive Decision Supported: Identifies amber-signal segments for enhanced active monitoring before they breach formal thresholds.

◆ Monitoring Priority Ranking

Business Question Answered: How should management rank the five monitored segments for resource allocation purposes?

Visual Selected: Horizontal bar chart with indexed priority scores (Modified Loans=100, High Risk Borrowers=80, Purchase Loans=60, New York=55, Serious Delinquency=40).

Alternatives Considered & Rejected: Ranked table (considered: a table would list the priority order clearly. Rejected: the bar chart's length encoding communicates the relative priority gaps more immediately than numbers in a table. The visual distance between Modified Loans (100) and Serious Delinquency (40) communicates relative urgency more powerfully). Donut chart (rejected: proportional area does not communicate priority ranking clearly).

Data Used: Hard-coded priority index values based on composite assessment of delinquency severity, exposure weight, and surveillance threshold proximity.

Measure Logic: Static indexed values assigned based on the analytical findings of Pages 1–5. The index values are not calculated dynamically — they represent the analyst's judgement-based priority ranking encoded as a consistent visual reference.

Key Insight: Modified Loans at 100 — clear priority leader. All others at 40–80 — active monitoring range.

Business Interpretation: The priority ranking translates the analytical findings into a resource allocation framework. When monitoring budgets are constrained, this chart answers where to concentrate oversight effort. The indexed format makes the ranking stable and comparable across reporting periods.

Executive Decision Supported: Operational answer to the resource allocation question: Modified Loans receives maximum oversight intensity, followed by High Risk Borrowers, then Purchase Loans and New York.

6 TECHNICAL DESIGN DECISIONS

Snapshot Model Implementation

The latest-snapshot fact table was built using a `row_number()` window function approach in PostgreSQL: partition by `loan_id`, order by `monthly_reporting_period` descending, and select only the first row per partition (`row_number = 1`). This produces exactly one record per loan — the most recent available observation — without requiring a self-join or subquery that would be impractical at 84 million row scale.

TECHNICAL NOTE *An alternative approach using `MAX(monthly_reporting_period)` per loan and then rejoining to the full table was considered. This produces identical results but requires two table scans — one for the `MAX` aggregation and one for the join. The `row_number()` approach requires one scan with a window function, which is more efficient at this scale.*

DAX Measure Design Philosophy

All KPI measures in the Power BI model were built to be filter-context aware by default. A delinquency rate measure applied in the context of a visual filtered to a specific state automatically returns the delinquency rate for that state — this is standard Power BI filter propagation behaviour. The design challenge is the KPI strip: those six measures must always return whole-portfolio figures regardless of which state, segment, or loan purpose filter is active on the page.

DESIGN DECISION *`REMOVEFILTERS` was applied to all six KPI strip measures to override page-level slicer context. The pattern used: `CALCULATE([Delinquency Rate], REMOVEFILTERS(Snapshot[property_state]), REMOVEFILTERS(Snapshot[risk_segment]), REMOVEFILTERS(Snapshot[credit_score_band]))`. This ensures that filtering to California on Page 2 does not alter the Total Portfolio Exposure or Portfolio Delinquency Rate cards at the top of the page.*

All other measures — those powering the analytical visuals — do not use `REMOVEFILTERS`. They are designed to respond to filter context, allowing the executive to filter to a specific state or segment and see all analytical charts update to reflect that filtered population while the KPI strip holds constant.

Conditional Formatting — Geographic Diagnostic Matrix

The Geographic Risk Diagnostic Matrix on Page 4 uses column-level conditional formatting based on portfolio averages as thresholds. Each metric column has its own conditional formatting rule: values above the portfolio average for that metric receive amber or red background colouring; values below receive green. The threshold for red versus amber was set at 1.5× the portfolio average for delinquency rate and modification rate metrics, and at 2× for High Risk borrower share.

The Total row in the matrix was intentionally formatted differently from the state rows — bold text only, no conditional colouring. The total row represents the portfolio average and serves as the benchmark against which individual state readings are evaluated; applying the same RAG colouring to the benchmark row itself would be logically circular.

Performance Optimisation

At 1.92 million rows in the Power BI import model, performance was adequate without additional optimisation for the analytical visuals. The primary performance consideration was the conditional-formatted matrix on Page 4, which calculates six separate metrics across ten state rows — 60 individual cell values. Pre-aggregating these values in a state-level summary measure rather than calculating at row context from the full snapshot table reduced visual rendering time.

For the full SQL analytical layer (84 million rows), the primary performance strategy was pre-aggregated summary tables. Running `GROUP BY` queries against the feature-engineered view directly was acceptable for single queries but impractical for iterative EDA. Creating permanent summary tables pre-calculating delinquency rates, exposure totals, and segment distributions reduced typical EDA query time from minutes to seconds.

Dashboard Navigation Design

Navigation is implemented through Power BI's built-in page tab structure, supplemented with a left sidebar icon panel that provides visual consistency across all six pages. Each sidebar icon corresponds to one dashboard page and is highlighted when that page is active. The sidebar was built using blank button shapes with custom fill colors and icon images rather than native page tabs — this approach allows the sidebar appearance to be consistent regardless of which page is active and provides cleaner visual alignment than native page tabs.

Monitoring Threshold Definition

The five monitoring thresholds used on Page 6 were set based on the analytical findings of Pages 1–5, not on external regulatory benchmarks. Modified Loans threshold of 20.0% was set slightly below the observed 22.8% — enough to flag the current position as a breach while setting a realistic improvement target. High Risk Borrowers threshold of 5.0% was set at approximately 2× the current 2.4% rate — a level that would represent material deterioration from the current position without setting an unrealistically tight limit. Similar logic applied to the remaining thresholds.

TECHNICAL NOTE *In a production institutional environment, monitoring thresholds would be formally approved by the Credit Risk Committee and documented in the Credit Risk Policy Framework. They would typically be calibrated against historical performance, stress test outcomes, and regulatory expectations. The thresholds in this project are analytically motivated but are not intended as substitutes for formally approved risk appetite limits.*

7 TECHNICAL LIMITATIONS

Snapshot Model Constraints

The latest-snapshot model is architecturally incapable of supporting any transition-based analysis. Roll rates, cure rates, and delinquency migration analysis all require observing the same loan's status at consecutive time points — which requires the full loan-month panel, not the snapshot. This is not a limitation that can be worked around within the snapshot model; it requires building additional analytical objects from the full panel dataset.

IMPORTANT LIMITATION *The inability to calculate roll rates and cure rates from the snapshot model is the most significant technical limitation of the dashboard's analytical scope. Roll rates are among the most important early warning indicators in mortgage portfolio surveillance — the rate at which 30 DPD loans transition to 60 DPD is a leading indicator of future serious delinquency that headline delinquency rates alone cannot provide. This capability exists in the SQL panel layer but was not surfaced in the dashboard.*

COVID-19 Measurement Suppression

The 2020 Q1–Q2 reporting period is the most policy-distorted period in US mortgage market history. Forbearance programme participation suppressed reported delinquency rates by an unknown but substantial amount — borrowers under active forbearance agreements were classified as current regardless of whether they were making payments. Foreclosure activity was near-zero due to the moratorium. The modification rate may itself be suppressed because formal modification was less necessary when forbearance provided an alternative relief mechanism.

The practical consequence for this project is that every delinquency metric understates true credit stress. The 0.65% delinquency rate represents a floor, not a ceiling. The analytical framework and surveillance infrastructure built in this project are sound; the specific threshold values calibrated against 2020 Q1–Q2 data will require recalibration as the post-forbearance performance picture emerges.

Origination-Only Variable Coverage

The key analytical variables — credit score, LTV, DTI — are available only as of origination. Credit scores change over time; a borrower who was 720 FICO at origination may have deteriorated to 640 by the time of this analysis. LTV changes as property values move; a loan originated at 80% LTV in 2018 may be at 70% or 90% LTV by 2020 depending on local house price movements. DTI changes as income and debt levels evolve. All three variables are used in this project as origination-period proxies, not as current measures of borrower financial condition.

No Macroeconomic Integration

The Fannie Mae dataset contains no macroeconomic variables. There is no unemployment rate, no house price index, no interest rate series embedded in the data. The geographic risk analysis identifies which states have elevated delinquency but cannot directly attribute that elevation to specific economic conditions. Integrating FHFA House Price Index data at the state level, or BLS unemployment rates at the MSA level, would enable the causal attribution step that this analysis cannot currently complete.

Conforming Conventional Scope

All findings are specific to the Fannie Mae conforming conventional mortgage segment. Government-backed loans (FHA, VA) — which historically have higher delinquency rates and serve a more credit-stressed borrower population — are not represented. Jumbo loans — which have stricter underwriting requirements and different risk profiles — are also excluded. The portfolio characteristics described in this project should not be extrapolated to the broader US mortgage market without segment-specific validation.

Why These Limitations Do Not Invalidate the Project's Conclusions

The limitations above define the boundary conditions of the analysis; they do not undermine its conclusions within those boundaries. The risk driver relationships identified — credit score gradient, risk layering effects, modification independence, geographic composition finding — are robust patterns observed across 1.92 million loans in a real institutional dataset. They are consistent with established academic and industry research on mortgage default risk. The analytical framework is sound; the specific quantitative values require periodic recalibration as reporting conditions normalise.

8 FUTURE ENHANCEMENTS

Roll Rate and Cure Rate Analysis

The highest-priority enhancement is building roll rate and cure rate analytics from the full 84.18 million row panel. This requires a self-join structure: each loan's delinquency status in month T is matched to its status in month T+1. Transition counts and probabilities can then be calculated for every DPD stage combination. The result is a roll rate matrix showing the monthly probability of each delinquency stage transitioning to each other stage — the foundational tool of mortgage portfolio stress testing.

Cure rate analysis — the probability of a delinquent loan returning to current status within N months — is the complementary metric. Both analyses exist in the SQL panel layer and could be developed as additional PostgreSQL views and imported into Power BI as supplementary tables to the current snapshot model.

Vintage Transition Analysis

Extending the panel analysis to track delinquency rates by origination vintage across multiple reporting periods would enable the vintage curve analysis referenced in the recommendations. The 2020 Q2 vintage shows low delinquency in the current snapshot — which was noted as likely reflecting insufficient seasoning rather than superior credit quality. Tracking this cohort forward through 2020 Q3–Q4 and 2021 would confirm or refute this interpretation.

IFRS 9 Expected Credit Loss Framework

The analytical building blocks for an IFRS 9 ECL framework are present in this project. EAD is available as current UPB per loan. PD can be estimated from the delinquency rate analysis — segment-level delinquency rates serve as PD proxies under the point-in-time approach. LGD requires recovery rate data from foreclosure exits, supplemented with industry benchmarks. Assembling these components into $ECL = PD \times EAD \times LGD$ per segment would produce a directional expected loss estimate by risk tier.

CREDIT RISK PERSPECTIVE *An IFRS 9-compliant ECL model requires three additional elements beyond what this project produces: (1) forward-looking macroeconomic adjustments to PD estimates, (2) staging logic (Stage 1: 12-month ECL; Stage 2: lifetime ECL for significant increase in credit risk; Stage 3: lifetime ECL for credit-impaired), and (3) discounting of future cash flow shortfalls at the effective interest rate. Developing these elements would transform the directional ECL estimate into a regulatory-grade calculation.*

Machine Learning Enhancement

A gradient boosting classifier (XGBoost or LightGBM) trained on the panel dataset to predict 90+ DPD within a 12-month forward window would complement the risk-layering segmentation approach with a continuous probability score. The categorical risk segment (Low/Moderate/High) is interpretable but loses discriminatory power within each tier. A continuous PD score would enable finer-grained loan-level ranking, particularly useful for collections outreach prioritisation within the Moderate Risk segment — where the delinquent exposure analysis identified the largest concentration of delinquent dollars.

Automated Monitoring and Alerting

The Portfolio Surveillance framework on Page 6 currently requires manual data refresh to update KPI values against thresholds. An automated monitoring pipeline — using Python or Power BI dataflow scheduling — could refresh the snapshot table on a defined cycle (monthly, matching the Fannie Mae reporting cadence) and automatically generate threshold breach alerts when a monitored segment crosses its limit. This would transform Page 6 from a periodic review tool into a continuous monitoring infrastructure.

Longitudinal Time-Series Analysis

The most analytically valuable future enhancement is extending the analysis across multiple quarterly datasets beyond 2020 Q1–Q2. The 2020 Q3–Q4 datasets would capture the post-forbearance delinquency emergence — the period when the COVID measurement suppression partially unwound and true borrower

stress became visible. Comparing pre-forbearance, during-forbearance, and post-forbearance delinquency rates across risk segments would provide the clearest possible evidence of which borrower characteristics predicted COVID-era default risk.

9 LESSONS LEARNED

SQL and PostgreSQL

Define the Grain Before Writing Any Query

The most consequential SQL lesson from this project: every analysis begins with a grain definition. Before writing any aggregation, the question must be answered: what does one row in my analytical result represent? For the panel, one row is one loan-month observation. For the snapshot, one row is one unique loan. For a summary table, one row might be one risk segment. Failing to define grain clearly at the outset produces aggregations that answer different questions than intended — and at 84 million row scale, the consequences of a wrong grain choice are not immediately visible.

Window Functions Are Essential for Panel Data

The `row_number()` window function approach used for snapshot creation is a pattern that applies across virtually any panel dataset transformation: selecting the first, last, or Nth observation per entity. Mastering window functions — `ROW_NUMBER`, `RANK`, `LAG`, `LEAD`, `FIRST_VALUE` — is more valuable than any individual optimisation technique for panel data analysis.

Pre-Aggregation Is a Production Requirement, Not a Shortcut

Running ad-hoc `GROUP BY` queries against 84 million rows for iterative EDA creates unworkable feedback cycles. Creating pre-aggregated summary tables — and rebuilding them when the underlying data changes — is standard practice in production analytics environments. The discipline of building this infrastructure at the start of EDA, rather than waiting until query times become frustrating, is a professional habit worth developing early.

Power BI and DAX

REMOVEFILTERS Discipline Prevents Reporting Errors

The most common Power BI error in multi-page dashboards is allowing page-level slicers to inadvertently affect KPI cards that are intended to show whole-portfolio metrics. Building the `REMOVEFILTERS` pattern into every KPI measure from the outset — rather than diagnosing and fixing filter propagation issues after the dashboard is built — saves significant rework and prevents silent reporting errors where the KPI cards show filtered values that executives misinterpret as portfolio-level facts.

Combo Charts Require Secondary Axis Management

The three combo charts in this project (Page 2: credit score band, Page 3: LTV band and loan purpose) all required careful secondary axis management to prevent one series from visually dominating the other. When loan count ranges from 10,000 to 1.13 million and delinquency rates range from 0.2% to 4.4%, placing both on the same axis makes one series nearly invisible. Secondary axis configuration is a mechanical step but its neglect produces misleading visuals.

Credit Risk Analytics

Exposure-Weighted Metrics Are Non-Negotiable

The recurring finding that the segment with the highest delinquency rate is not the segment with the most delinquent exposure reinforced a lesson that applies universally in credit risk analytics: delinquency rates tell you where risk is most intense; exposure-weighted metrics tell you where risk is most financially consequential. Any portfolio monitoring framework that reports only rates will systematically misallocate oversight resources. Always calculate both.

Independence Testing Changes the Analytical Conclusion

The modification status finding on Page 5 — that modification carries independent risk above and beyond risk segment classification — was only discoverable through the cross-segment analysis (Page 5's clustered bar chart). If the analysis had stopped at the headline 22.8% vs 0.6% delinquency comparison, the

conclusion would have been 'modified loans are high risk.' The independence test changed the conclusion to 'modification status provides predictive value that risk segment classification does not capture' — a materially different and more actionable finding. Always test whether significant findings are independent of existing classification variables.

Dashboard Design

Diagnostic Pages Are More Valuable Than Ranking Pages

The most impactful design decision across the entire project was reframing Page 4 from 'which states are risky' to 'why are certain states riskier.' A ranking page answers a question executives can partially answer from general knowledge. A diagnostic page answers a question that requires the analysis to answer — and therefore delivers genuine analytical value. Every dashboard page should pass the test: 'Can the executive answer this question without this analysis?' If yes, the page may be a ranking page masquerading as a diagnostic page.

Design Constraints Improve Quality

The rule that each page must answer exactly one executive business question and that no visual may appear on a page unless it directly contributes to that question was initially restrictive and ultimately clarifying. Several visuals that were analytically interesting were excluded from their intended pages because they answered a slightly different question. The constraint forced prioritisation and produced a more coherent dashboard than unconstrained visual addition would have.

10 CONCLUSION

This project began with 84.18 million rows of raw Fannie Mae mortgage performance data — pipe-delimited, header-free, and requiring complete field mapping before any analytical work could begin. It concluded with a six-page executive dashboard, a formal Executive Credit Risk Report, a Business Case Study, and this Technical Documentation — a complete analytical and communication ecosystem built on top of a production-scale real institutional dataset.

The technical journey required deliberate architectural decisions at every stage. The choice to load all 108 columns as TEXT and apply type casting in a downstream view — rather than attempting typed ingestion — prevented ingestion failures on a dataset of this scale. The choice to build the feature engineering as a view on top of the typed view — rather than a permanent table — allowed iterative feature refinement without rebuilding the full dataset. The choice to create pre-aggregated summary tables before beginning EDA — rather than running repeated full-table queries — made the analytical workflow practical at 84 million row scale.

The most significant technical decision was the creation of the latest-snapshot fact table as a separate analytical layer for Power BI reporting. This decision required understanding the difference between what the data shows (cumulative monthly panel history) and what executives need to see (current portfolio position) — and building the infrastructure to produce the latter from the former without compromising either analytical layer. The result is a dashboard whose KPIs accurately reflect current portfolio status while the SQL layer retains the full historical record for longitudinal analysis.

The dashboard design decisions were equally deliberate. One executive business question per page. Consistent KPI strip across five of six pages. Diagnostic reframing of the geographic analysis. Three-visual minimalism on Page 5 to let the strongest finding speak clearly. Mode change to operational monitoring on Page 6. Each decision was made because it produced a better analytical communication product — not because it was technically convenient or visually interesting.

The outcome is a project that functions simultaneously at multiple levels. As an executive communication tool, it provides a CRO with a structured path from portfolio health through to operational monitoring priorities in six pages. As a technical portfolio project, it demonstrates end-to-end credit risk analytics capability across SQL engineering, data modelling, feature engineering, dashboard development, and professional reporting. As an interview preparation resource, it provides documented, reasoned answers to every technical question a credit risk analytics interviewer might raise.

Final statement: *This project demonstrates that credit risk analytics is not primarily a technical discipline — it is a business communication discipline that requires technical precision to be credible. The PostgreSQL engineering, the data modelling philosophy, the DAX measure design, and the dashboard architecture all serve one purpose: enabling the right decision to be made by the right person with the right information. That alignment of technical rigour with business purpose is what distinguishes analytical work that creates institutional value from analytical work that simply produces output.*

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